

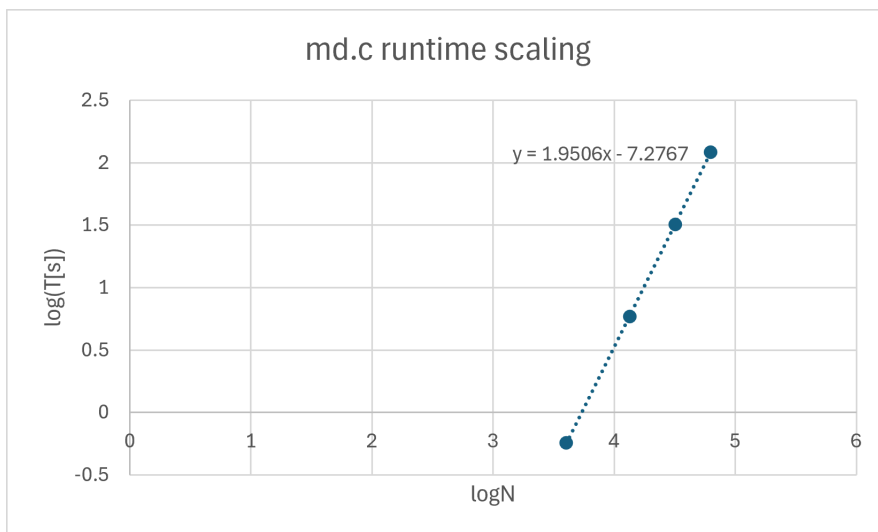
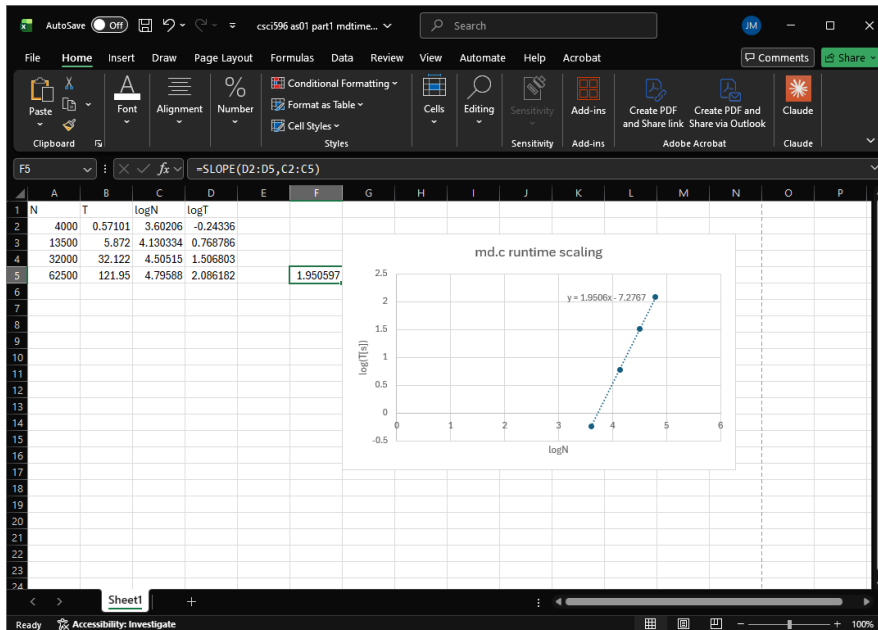
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CSCI596 Assignment 1—Computational Complexity and Flop/s Performance

1. Measuring Computational Complexity

Submit (i) the plot and (ii) your fitted value of α .

(i) my log-log plot of number of atoms (N) and runtime in seconds (T):



(ii) my fitted value of α from the linear fit $\log T = \alpha \cdot \log N + \beta$

$\alpha = 1.9506$

2. Theoretical Flop/s Performance

What is the theoretical peak performance of your computer in terms of double-precision Gflop/s (gigaflop/s or 10⁹ flop/s)? Submit the computed number.

#processors | 1
#cores_per_processors | 8
clock_speed [GHz] | 2.3
#FMA_units | 1
vector_size [bits] | 256

The theoretical flop/s of a computer is computed as

Peak_flop/s = (#processors) × (#cores_per_processor) × (clock_speed [1/s]) × (2 × #FMA_units) × (vector_size [bits] / 64)

= 1 × 8 × 2.3 × (2 × 1) × (256 / 64)

= 147.2 Gflop/s

(Optional) Compile and run the lmd_sqrt_flop.c program on a computer of your choice, and report the flop/s performance you get. Better, answer how many % of the theoretical peak flop/s performance of the computer you achieved?

\$ gcc -O -o lmd_sqrt_flop lmd_sqrt_flop.c -lm
./lmd_sqrt_flop < lmd_sqrt_flop.in
Execution time (s) = 3.574700e-02
Number of FP operations = 1.609487e+08
MFlops rate = 4.502440e+03

Theoretical of 1 core of my processor

Peak_flop/s = 1 × 1 × 4.3 × (2 × 2) × (512 / 64)
= 137.6 Gflop/s

4.50244 / 137.6 = 0.03272

≈ 3.3% of the theoretical peak flop/s